Assignment set IV

Wednesday, July 8, 2026 9:58 AM

1.1) lorenz system

$$\begin{cases} \dot{x} = -\sigma x + \sigma y \\ \dot{y} = rx - y - xz \\ \dot{z} = -bz + xy \end{cases} \qquad L_3(x,y,z) = \frac{rx^2 + \sigma y^2 + \sigma(z-2r)^2}{2}$$

(a)

$$L_3' = rx\dot{x} + \sigma y\dot{y} + \sigma\dot{z}(z-2r) = rx(-\sigma x + \sigma y) + \sigma y(rx - y - xz) + \sigma(z-2r)(-bz + xy) =$$

$$= -r\sigma x^2 + r\sigma xy + r\sigma xy - \sigma y^2 - \sigma xyz - \sigma bz^2 + \sigma xyz + 2r\sigma bz - 2r\sigma xy =$$

$$= -r\sigma x^2 - \sigma y^2 - \sigma bz^2 + 2r\sigma bz = \sigma br^2 - \sigma\left(b(z-r)^2 + rx^2 + y^2\right)$$

(b) $L' = \sigma br^2 - \sigma\left(b(z-r)^2 + rx^2 + y^2\right) \geq 0$

$$br^2 \geq b(z-r)^2 + rx^2 + y^2$$

so for $(x,y,z) \in D = \{(x,y,z) \mid b(z-r)^2 + rx^2 + y^2 \leq br^2\} \iff L'(x,y,z) \geq 0$

so within this ellipsoid, $L' \geq 0$, out of it $L' < 0$

for $(x,y,z) \in D$, since $b(z-r)^2 + rx^2 + y^2 \leq br^2$

we see - $b(z-r)^2 \leq br^2$

$rx^2 \leq br^2$ (I)

$y^2 \leq br^2$ (II)

since $(z-r)^2 \leq r^2$

$z^2 - 2rz \leq 0$

$0 \leq z^2 \leq 2rz$

we can say $0 \geq z^2 - 2rz \geq z^2 - 4rz$

$4r^2 \geq z^2 - 4rz + 4r^2 = (z-2r)^2$

therefor $(z-2r)^2 \leq 4r^2$ (III)

so for any $(x,y,z) \in D$, $L(x,y,z) = \frac{1}{2}\left(rx^2 + \sigma y^2 + \sigma(z-2r)^2\right) \leq \frac{1}{2}(br^2 + \sigma br^2 + 4\sigma r^2)$ (I) (II) (III)

$$= \boxed{\frac{r^2}{2}(b + \sigma b + 4\sigma) = c_3}$$

so $(x,y,z) \in D \Rightarrow L(x,y,z) \leq c_3$

and $(x,y,z) \notin D \Leftarrow L(x,y,z) > c_3$

so $(x,y,z) \in V \Rightarrow L(x,y,z) \le c_3$

and $(x,y,z) \notin O \Leftarrow L(x,y,z) > c_3$

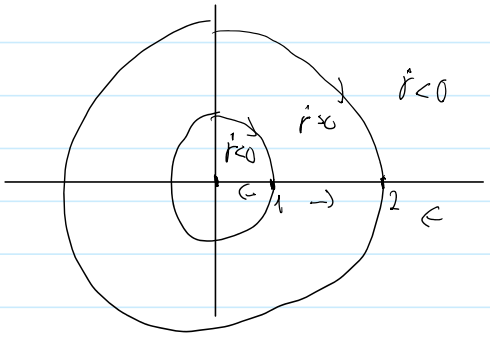
we saw before that $(x,y,z) \in O \Leftrightarrow L' \ge 0$

then we get $L(x,y,z) > c_3 \Rightarrow L' < 0$

(c)

1.2) $\dot{r} = r(1-r^2)(r^2-4)$

$\dot{\theta} = 1$



(a) on $r=1$ and $r=2$ we have periodic orbit

lets analyze $\dot{r}$ at $0<r<1$, $1<r<2$, $r>2$

$0<r<1$: $\dot{r} = \overset{+}{r}(1-\overset{+}{r^2})(\overset{-}{r^2}-4) < 0$

$1<r<2$: $\dot{r} = \overset{+}{r}(1-\overset{-}{r^2})(\overset{-}{r^2}-4) > 0$

$r>2$: $\dot{r} = \overset{+}{r}(1-\overset{-}{r^2})(\overset{+}{r^2}-4) < 0$

for any $x_0 = (r_0, \theta) \in \{r=1\}$ take $d = \frac{1}{2}$, $\forall \delta > 0$ neighborhood around x_0, we can take $V_0 = (r', a)$ where $r' < 1$ since $r < r_0$ those orientations $\varphi(t, V_0)$ will go to $r=0$ while

for any $X_0 = (r_0, \theta) \in \{r=1\}$ take $d = \frac{1}{2}$, $\forall \delta > 0$ neighborhood around X_0, we can take $Y_0 = (r_0', \theta)$, where $r_0' < 1$. since $\dot{r} < 0$ there, eventually $\varphi(t, Y_0)$ will go to $r=0$, while $\varphi(t, X_0)$ will stay at $r=1$. so $\|\varphi(t, X_0) - \varphi(t, Y_0)\| \xrightarrow{t\to\infty} 1 > d = \frac{1}{2}$

so $\{r=1\}$ has sensitive dependence on initial condition.

for any point $(r_0, \theta) = X_0 \in \{r=2\}$ and any $Y_0 = (r_0', \theta)$ in any δ so neiborhood of X_0

Case 1: $r_0' \le 2$, $\dot{r} < 0$ so it spirals to $r=2$

Case 2: if $Y_0 = 2$, then its on the orbit and it will always orbit with X_0

Case 3: if $Y_0 < 2$, $\dot{r} > 0$, again, it spirals towards $r=2$

since $\dot{\theta} = 1$, any X_0, Y_0 spin in the same angular motion, so they can't get far in that way, and since for all Y_0, $r_0' \xrightarrow{t\to\infty} 2$ monotonically

for any $d > 0$ you pick, eventually $\|\varphi(t, X_0) - \varphi(t, Y_0)\| < d$

so $\{r=2\}$ does not have sensitive dependence on initial condition.

(b) for the system to have sensitive dependence on initial condition when restricted to either $\{r=1\}$ or $\{r=2\}$, means the Y_0 we pick has to be from these sets.

in both case if both X_0 and Y_0 are on the orbit, their $\dot{r} = 0$ and $\dot{\theta} = 1$ and this will remain true forever. so for any $d > 0$ we pick, we can take $\delta < d$ so Y_0 will be close enough and $\forall t$ $\|\varphi(t, X_0) - \varphi(t, Y_0)\| = \|\varphi(0, X_0) - \varphi(0, Y_0)\| < d$ since they will rotate together.

(c) neither $\{r=1\}$ nor $\{r=2\}$ are chaotic attractors, since they are not sensitive when restricted,

1.3)

$$\dot{x} = -\mu x + yz$$
$$\dot{y} = -\mu y + (z - A)x$$
$$\dot{z} = 1 - xy$$

(a) $\dot{z} = 1 - xy = 0$

$y = \frac{1}{x}$

$\dot{x} = -\mu x + yz = 0$

$-\mu x + \frac{z}{x} = 0$

$z = \mu x^2$

$\dot{y} = -\mu y + (z - A)x = 0$

$-\frac{\mu}{x} + \mu x^3 - Ax = 0$

$x^2 - x^{-2} = \frac{A}{\mu}$

$z - \mu x$

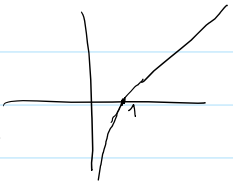
$x^2 - x^{-2} = \frac{A}{\mu}$

let's look at $x^2 = a$, $f(a) = a - \frac{1}{a}$ for $a > 0$

$f'(a) = 1 + \frac{1}{a^2} > 0$

so for $\frac{A}{\mu} > 0$ there is exactly one a^* where $f(a^*) = \frac{A}{\mu}$

(assuming $A, \mu > 0$)



so $x = \pm\sqrt{a^*} = \pm\alpha$

the FP are - $(x_1^* y_1^* z_1^*) = (\alpha, \frac{1}{\alpha}, \mu\alpha^2)$ and $(x_1^* y_1^* z_1^*) = (-\alpha, -\frac{1}{\alpha}, \mu\alpha^2)$

(b) from (a), at FP $A = \mu(x^2 - x^{-2})$, so $z - A = \mu x - \mu(x^2 - x^{-2}) = \frac{\mu}{x^2}$

$$Df(x,y,z) = \begin{pmatrix} -\mu & z & y \\ \frac{\mu}{x^2} & -\mu & x \\ -y & -x & 0 \end{pmatrix}$$

$$Df(x_1^*, y_1^*, z_1^*) = \begin{pmatrix} -\mu & \mu\alpha^2 & \frac{1}{\alpha} \\ \frac{\mu}{\alpha^2} & -\mu & \alpha \\ -\frac{1}{\alpha} & -\alpha & 0 \end{pmatrix} = A_1$$

$$\det(A_1 - I\lambda) = \left| \begin{pmatrix} -\mu-\lambda & \mu\alpha^2 & \frac{1}{\alpha} \\ \frac{\mu}{\alpha^2} & -\mu-\lambda & \alpha \\ -\frac{1}{\alpha} & -\alpha & -\lambda \end{pmatrix} \right| =$$

$$= (-\mu-\lambda)(\mu\lambda + \lambda^2 + \alpha^2) - \mu\alpha^2\left(-\frac{\mu\lambda}{\alpha^2} + 1\right) + \frac{1}{\alpha}\left(-\frac{\mu}{\alpha} - \frac{\mu}{\alpha} - \frac{\lambda}{\alpha}\right) =$$

$$= -\mu^2\lambda - \mu\lambda^2 - \mu\alpha^2 - \mu\lambda^2 - \lambda^3 - \alpha^2\lambda + \mu^2\lambda - \mu\alpha^2 - \frac{2\mu+\lambda}{\alpha^2} =$$

$$= -(\lambda^3 + 2\mu\lambda^2 + 2\mu\alpha^2 + \alpha^2\lambda + \frac{2\mu}{\alpha^2} + \frac{\lambda}{\alpha^2}) = 0$$

$A = \mu(\alpha^2 - \alpha^{-2})$ ✓ (i think it was meant to be $A = \mu(\alpha^2 + \alpha^{-2})$)

$$\frac{A}{\mu}\lambda + 2A = \lambda\alpha^2 - \frac{\lambda}{\alpha^2} + 2\mu\alpha^2 - \frac{2\mu}{\alpha^2}$$

✓ ✓ ✓ -

if it was $\lambda^3 + 2\mu\lambda^2 + \frac{A}{\mu}\lambda + 2A = (\lambda + 2\mu)(\lambda^2 + \frac{A}{\mu}) = 0$

if it was $\lambda^3 + 2\mu\lambda^2 + \frac{A}{\mu}\lambda + 2A = (\lambda + 2\mu)\left(\lambda^2 + \frac{A}{\mu}\right) = 0$

$\lambda = -2\mu$ $\qquad$ $\lambda^2 = -\frac{A}{\mu}$

$\lambda = \mp i\sqrt{\frac{A}{\mu}}$

(c) for a region U, $\varphi(t;U)$ is where that region move after t
$V(t)$ is the volume of $\varphi(t;U)$

as we saw in class, $\dot{V} = \int_U \sum_i \frac{\partial F_i}{\partial x_i}$

in our case $\dot{V} = \int_U \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{y} + \frac{\partial \dot{z}}{z} = \int_U -2\mu = -2\mu \int_U 1 = -2\mu \cdot V$

so $\dot{V} = -2\mu V$

since V is a volume $V \geq 0$ so $\dot{V} < 0$, the total system volume decreases with time